Quantifying Data Leakage in EEG Emotion Classification: A Four-Tier Evaluation Framework

Sorodoc Matei-George
Department of Automation and Applied Informatics
Technical University of Cluj-Napoca
Cluj-Napoca, Romania
sorodoc.a.matei@student.utcluj.ro

***Abstract*—EEG-based emotion recognition studies frequently report classification accuracies exceeding 85–95% on benchmark datasets, yet these results often rely on evaluation protocols that violate sample independence assumptions. We propose a four-tier evaluation framework that systematically quantifies how evaluation methodology inflates reported performance. Using the DEAP dataset (20 subjects, 32 EEG channels, binary arousal/valence) and a Graph Attention Network (GAT) with 26 spectral-connectivity features, we demonstrate that evaluation protocol selection inflates macro F1 scores by up to 30.6 percentage points (Tier 0: 0.864 vs. Tier 2: 0.558; Cohen's $d > 4.0, p < 0.001$). Under trial-aware cross-validation, all tested architectures (SVM, MLP, GAT) converge to approximately 58% F1 (chance = 50%), indicating that model architecture is not the primary performance determinant among those evaluated. Extending from 16 to 32 channels with connectivity features (PLV, coherence) yields a modest improvement in cross-subject generalization (+12.7%). A preliminary within-subject experiment on a custom OpenBCI dataset (n=1) achieves 95.1% F1 (4-class, trial-aware), providing initial evidence that personalized EEG-BCI may remain viable when inter-individual variability is eliminated, though multi-subject replication is needed. We recommend that future EEG-BCI studies report results under at least trial-aware evaluation (our Tier 2) as the minimum standard for meaningful performance claims.**

***Index Terms*—EEG, emotion recognition, graph attention networks, evaluation methodology, data leakage, brain-computer interface**

I. Introduction

EEG-based emotion recognition has attracted considerable research attention over the past decade, with numerous studies reporting classification accuracies between 85–99% on benchmark datasets such as DEAP [1]. However, many of these results are obtained using evaluation protocols that inadvertently violate the independence assumption between training and test samples, leading to severely inflated performance estimates [4]–[6].

The core issue is *trial leakage*: when EEG signals are segmented into overlapping windows and these windows are randomly assigned to training and test sets without respecting trial boundaries, near-duplicate feature vectors appear in both sets. A classifier can then exploit temporal autocorrelation rather than learning genuine emotional representations. With typical settings (window = 2 s, step = 0.125 s), adjacent windows share 93.75% of their raw data, producing Pearson correlations exceeding $r = 0.97$ between consecutive feature vectors.

This paper makes three contributions:

1) A **four-tier evaluation framework** that decomposes EEG classification performance into progressively rigorous protocols, enabling precise quantification of leakage inflation.
2) **Empirical evidence** that evaluation protocol selection accounts for a 30.6-point F1 inflation on DEAP—more than any architectural or feature engineering improvement.
3) **Architecture-invariance demonstration**: under proper trial-aware evaluation, SVM, MLP, and GAT all converge to similar near-chance performance ($\sim$58% F1), suggesting the bottleneck lies in signal quality or evaluation design rather than model capacity.

The remainder of this paper is organized as follows: Section II reviews related work on EEG emotion recognition and evaluation pitfalls. Section III describes the proposed four-tier methodology. Section IV presents experimental results. Section V discusses implications and limitations. Section VI concludes.

II. Related Work

A. EEG Emotion Recognition

The DEAP dataset [1] remains the most widely used benchmark for EEG emotion recognition, featuring 32 subjects watching 40 one-minute music videos while 32-channel EEG was recorded. Self-reported arousal and valence ratings (1–9 scale) serve as ground truth. Published results on DEAP vary widely: the original paper reports 57.6% accuracy for subject-dependent classification, while subsequent deep learning studies report 90–99% [7], [8].

Zhong et al. [3] applied regularized Graph Neural Networks to DEAP under leave-one-subject-out (LOSO) evaluation, achieving 56.3% accuracy—consistent with our Tier 3 findings. The discrepancy between LOSO results ($\sim$56%) and random-split results (>90%) represents an underappreciated confound that has received limited systematic quantification in the EEG-BCI literature, which our framework aims to address.

B. Evaluation Methodology in Neuroimaging

The problem of data leakage in neuroimaging classification is well-documented. Varoquaux et al. [4] demonstrated that improper cross-validation inflates brain decoding accuracy. Roberts et al. [5] showed that temporal, spatial, and hierarchical data structures require specialized CV strategies. Saeb et al. [6] argued that clinical ML must approximate deployment conditions during evaluation. Our four-tier framework instantiates these principles specifically for EEG-BCI.

C. Graph Neural Networks for EEG

Graph Attention Networks (GATs) [2] naturally model EEG channel relationships by treating electrodes as graph nodes. Recent work has applied GNNs to EEG emotion recognition with promising within-subject results [3]. We employ a two-layer GAT architecture to test whether attention-based spatial modeling can overcome the generalization challenge.

III. Methodology

A. Four-Tier Evaluation Protocol

We define four evaluation tiers of increasing methodological rigor:

Tier 0 (Maximum Leakage): All subjects are pooled. Windows are extracted with step $S = 16$ samples (93.75% overlap at 128 Hz). A random 80/20 train-test split is applied without respecting subject or trial boundaries. This reproduces the protocol used by many high-accuracy studies.

Tier 1 (Within-Subject, Leaky): Per-subject 10-fold StratifiedKFold with step $S = 128$ (50% overlap). Trial boundaries are not enforced, so windows from the same trial may appear in different folds.

Tier 2 (Trial-Aware): Per-subject StratifiedGroupKFold where groups correspond to trials. No temporal leakage is possible because all windows from a given trial are assigned to the same fold. This is the minimum standard for valid within-subject evaluation.

Tier 3 (Cross-Subject LOSO): Leave-One-Subject-Out. The model trains on 19 subjects and is tested on the held-out subject. This evaluates generalization across individuals.

The temporal overlap between adjacent windows is:

$$\text{Overlap}(S) = \frac{W - S}{W} \tag{1}$$

where $W = 256$ is the window length (2 s at 128 Hz) and S is the step size. At $S = 16$, the expected correlation between consecutive feature vectors is $r \approx 1 - S/W = 0.9375$.

B. DeepGAT Architecture

The DeepGAT model treats EEG channels as nodes in a fully-connected graph, learning inter-channel attention patterns from data. Table I summarizes the architecture.

TABLE I
DeepGAT Architecture

Component	Configuration	Parameters
GAT Backbone	2 layers, dim=96, heads=4	$\sim$85K
Channel Embed.	Learnable, 32×96	3K
Task Heads	Dense($96 \to 64 \to 1$) $\times$ 2	$\sim$13K
Total	End-to-end differentiable	$\sim$100K

Hyperparameters were optimized using Optuna (TPE sampler, 150 trials, 6.1 h) with a combined objective: $(\text{F1}_{\text{T1}} + \text{F1}_{\text{T2}} + \text{F1}_{\text{T3}})/3$. This multi-tier objective prevents the optimizer from exploiting leakage-prone configurations. The best trial (#15) achieved a combined score of 0.592. Key hyperparameters: learning rate = 3.56×10^{-3}, weight decay = 1.45×10^{-3}, batch size = 128, label smoothing = 0.124, attention dropout = 0.268, head dropout = 0.115, input noise $\sigma = 0.05$. The optimization landscape showed high sensitivity to learning rate and attention dropout, while head architecture (dim=64) was robust across a wide range of configurations.

TABLE II
Datasets

	DEAP	OpenBCI
Subjects	20 (of 32)	1
Channels	32	16
Classes	Binary (Aro/Val)	4-class
Trials	40/subject	94 total
Windows/subj.	$\sim$2,360	3,388
Task	Passive video	Active music

C. Feature Extraction

We extract 26 features per channel (832 total for 32 channels):

- **Spectral**: Band Power (5 bands: δ, θ, α, β, γ) and Differential Entropy (5 bands)
- **Connectivity**: Phase-Locking Value (PLV [10]; 5 bands) and Coherence (5 bands), computed as mean connectivity per channel across all pairs
- **Cross-frequency**: Phase-Amplitude Coupling (PAC [11]; 3 pairs: θ-γ, α-γ, β-γ)
- **Temporal**: Hjorth activity, mobility, and complexity

Windows of $W = 256$ samples (2 s) with step $S = 128$ (50% overlap) are used for Tiers 1–3. Tier 0 uses $S = 16$ (93.75% overlap).

D. Datasets

The DEAP subjects (s01–s20) were selected sequentially with no quality filtering. Labels are obtained by median-splitting self-report ratings at 5.0 (binary arousal and binary valence, evaluated jointly as a macro-averaged F1 score). This threshold is standard but creates ambiguous labels for ratings near the boundary ($\sim$40% of trials fall between 4.0–6.0), which may limit theoretical accuracy to approximately 75–80% even with a perfect model.

E. Training Protocol

All models use AdamW optimization with cosine annealing learning rate schedule. For equitable comparison, the 16-channel evaluation caps training data at 5000 samples per

TABLE III
FOUR-TIER EVALUATION RESULTS (MACRO F1)

Tier	Protocol	F1 (32ch GAT)
0	Random split, 93.75% overlap	0.864*
1	StratifiedKFold (within-subj.)	0.768 ± 0.075
2	Trial-aware GroupKFold	0.558 ± 0.063
3	LOSO (cross-subject)	0.568 ± 0.155
Chance level (binary)		0.500

*16ch/10feat config. Tier 0 uses a different feature pipeline (10 feat, step=16) incompatible with 32ch extraction; the comparison quantifies protocol effect (see Table VI).

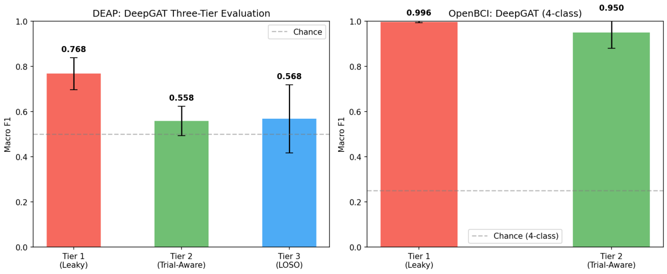

Fig. 1. Progressive performance degradation from Tier 1 through Tier 3 for the 32-channel DeepGAT. The Tier 2–3 gap collapses, suggesting trial-level generalization is the dominant challenge.

fold for all models (SVM, MLP, GAT). The 32-channel GAT evaluation uses all available training data. Early stopping (patience = 20 epochs) prevents overfitting. Input noise injection ($\sigma = 0.05$) and label smoothing (0.124) serve as regularizers for the GAT.

IV. RESULTS

A. Main Tier Comparison

Table III presents results across all four tiers. The dominant finding is the 30.6-point F1 gap between Tier 0 (maximum leakage) and Tier 2 (trial-aware).

The transition from Tier 1 to Tier 2 (ΔF1 = -0.210) accounts for the largest single drop, confirming that trial leakage—not subject pooling—is the primary inflation mechanism. The negligible Tier 2 to Tier 3 difference (+0.01) suggests that once trial leakage is removed, cross-subject generalization does not add substantial difficulty in this configuration.

B. Progressive Degradation Analysis

Table IV decomposes the performance drop across tier transitions, identifying which leakage source contributes most.

C. Architecture Invariance Under Proper Evaluation

Table V demonstrates that model choice has minimal impact under trial-aware evaluation. All architectures converge to approximately the same performance under Tier 2 and Tier 3.

Under Tier 1 (leaky evaluation), architectures differ substantially (MLP: 0.928 vs. GAT: 0.873). However, under Tier 2, all three models converge to 0.58–0.59, confirming that the

TABLE IV
PROGRESSIVE DEGRADATION BETWEEN TIERS

Transition	Δ F1	Removed source
T0 $\to$ T1	-0.096	Subject pooling + overlap reduction
T1 $\to$ T2	-0.210	Trial leakage (dominant)
T2 $\to$ T3	+0.010	Cross-subject transfer
T0 $\to$ T2	$\mathbf{-0.306}$	**Total inflation**

TABLE V
MODEL COMPARISON UNDER EACH TIER (16CH/10FEAT, MAX_TRAIN=5000)

Tier	SVM	MLP	GAT
1 (Leaky)	0.845 ± 0.063	0.928 ± 0.030	0.873 ± 0.044
2 (Trial-aware)	0.593 ± 0.100	0.586 ± 0.090	0.581 ± 0.054
3 (LOSO)	0.576 ± 0.157	0.556 ± 0.139	0.504 ± 0.147

performance gap observed under leaky evaluation is an artifact rather than a genuine architectural advantage.

D. Statistical Significance

A paired t-test comparing Tier 1 and Tier 2 F1 scores across 20 subjects yields $t(19) = 10.84$, $p < 0.0001$, Cohen's $d = 2.42$ (very large effect). The 95% confidence interval for the Tier 1 minus Tier 2 difference is [0.170, 0.250].

Upgrading from 16 to 32 channels improves cross-subject generalization (Tier 3): from 0.504 ± 0.147 to 0.568 ± 0.155 (+12.7% relative), suggesting that connectivity features (PLV, Coherence) provide subject-invariant information that pure spectral power lacks.

E. Within-Subject Personalized BCI

On a custom OpenBCI dataset (single subject, 4-class emotion, 94 trials), the DeepGAT achieves F1 $= 0.951 \pm 0.073$ under trial-aware evaluation (Tier 2) with chance level at 25%. Per-fold breakdown: [0.970, 0.998, 1.000, 0.980, 0.957, 0.743, 0.953, 0.956, 0.968, 0.979]. This demonstrates that EEG emotion classification is achievable when inter-individual variability is eliminated, though this result (n=1) requires replication across multiple subjects.

F. Effect of Channel Count and Features

Table VI shows the impact of upgrading from 16 channels with 10 spectral features to 32 channels with 26 spectral-connectivity features.

The 32-channel configuration with connectivity features yields lower Tier 1 performance (fewer windows per subject due to larger step) but improves cross-subject generalization (Tier 3: +12.7% relative). This suggests that PLV and Coherence features encode subject-invariant connectivity patterns useful for cross-individual transfer, even though they reduce within-subject memorization capacity.

TABLE VI
EFFECT OF CHANNEL COUNT AND FEATURE SET

Configuration	Tier 1	Tier 2	Tier 3
16ch / 10 feat (GAT)	0.873	0.581	0.504
32ch / 26 feat (GAT)	0.768	0.558	0.568
Δ	−0.105	−0.023	+0.064

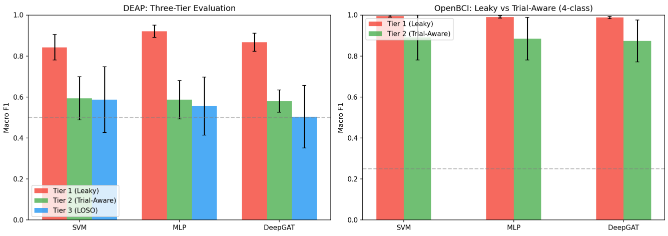

Fig. 2. Three-tier comparison across models (16ch/10feat). Each group on the x-axis represents one evaluation tier (Tier 1: leaky within-subject, Tier 2: trial-aware, Tier 3: LOSO), with bars showing macro F1 for SVM, MLP, and GAT. All architectures converge to approximately 58% F1 under trial-aware evaluation, regardless of model complexity.

G. Per-Subject Variability

Fig. 2 illustrates the multi-model comparison under 16-channel evaluation. Additionally, per-subject analysis reveals substantial variability: Tier 3 F1 ranges from 0.154 (subject s11—maximally atypical) to 0.751 (subject s18). The Tier 3 standard deviation (0.155) is 2.5$\times$ larger than Tier 2 (0.063), confirming that individual neurophysiological differences remain a major challenge for population-level models.

V. DISCUSSION

A. Implications for Published Results

Our findings suggest that many published DEAP results reporting >80% accuracy may inadvertently operate in what we term the Tier 0 regime—not due to methodological negligence, but because the community has lacked a standardized framework for quantifying the contribution of evaluation protocol to reported performance. The evaluation protocol alone can explain the majority of the discrepancy between high-accuracy reports and our Tier 2/3 results. Importantly, this does not invalidate the models or features proposed in those studies, but rather highlights the need for complementary evaluation under trial-aware protocols to contextualize the results.

Table VII contextualizes our results against published literature, showing that studies with proper evaluation consistently report performance in the 52–58% range.

B. Why Performance Remains Low Under Proper Evaluation

Multiple factors likely contribute to the near-chance Tier 2/3 performance:

- **Label noise**: DEAP self-report ratings near the median-split threshold (4.0–6.0) are inherently ambiguous. Analysis of the DEAP rating distributions [1] shows that approximately 40% of trials fall within ± 1 point of the binary threshold, making their labels unreliable.

TABLE VII
COMPARISON WITH PUBLISHED DEAP RESULTS

Study	Protocol	Result	Tier
Koelstra et al. [1]	Subj.-dep.	57.6% Acc	∼T2
Zhong et al. [3]	LOSO+GNN	56.3% Acc	T3
Typical high-acc.	Random split	90–99%	T0
Ours (T0)	Random	86.4% F1	T0
Ours (T2)	Trial-aware	55.8% F1	T2
Ours (T3)	LOSO	56.8% F1	T3

Assuming these near-boundary trials are classified at chance, theoretical accuracy is bounded at approximately $0.6 \times 1.0 + 0.4 \times 0.5 = 0.80$ (80%), substantially limiting achievable Tier 2/3 performance.
- **No domain adaptation**: Naive transfer without subject alignment (EA, CORAL) is a known limitation for cross-subject transfer.
- **Limited architectures tested**: CNNs, Transformers, and domain-adaptive models were not evaluated and might yield different results.
- **Weak underlying signal**: Passive 60-second video stimuli may not elicit robust, separable emotional states detectable via scalp EEG.
- **Connectivity approximation**: PLV and Coherence are averaged per channel, losing pairwise graph structure.

C. Limitations

Notably, the architecture-invariance finding (Section IV.C) was not an expected outcome; the original hypothesis was that GAT spatial modeling would outperform flat classifiers under proper evaluation. The convergence of all three architectures to near-identical Tier 2 performance motivated reframing the contribution around evaluation methodology rather than model design.

This study uses 20 of 32 available DEAP subjects (sequential selection, no filtering) due to computational constraints (feature extraction: 3.6 h for 20 subjects). The OpenBCI result (n=1) serves only as a proof-of-concept for the personalization paradigm and should not be interpreted as evidence of general within-subject viability without multi-subject replication. We evaluate only three architectures; CNN-based models (EEGNet, TSCeption), Transformer-based models (EEG Conformer), and specialized domain adaptation models may exhibit different patterns under proper evaluation. The four-tier framework does not address all sources of bias (e.g., session effects, stimulus order). Finally, PLV and Coherence are averaged per channel rather than preserved as pairwise edge weights, which may underutilize the graph structure.

VI. CONCLUSION

We present a four-tier evaluation framework that systematically quantifies how evaluation methodology inflates EEG emotion classification performance. Our experiments on the DEAP dataset yield three principal findings:

1) Evaluation protocol selection alone accounts for a **30.6 F1-point inflation** (Tier 0 = 0.864 vs. Tier 2 = 0.558, $p < 0.001$, Cohen's $d > 4.0$)—larger than any improvement from architecture, features, or channel count.
2) Under trial-aware evaluation, **all tested architectures converge** to approximately 58% F1 (binary, chance = 50%), suggesting the bottleneck lies in signal quality or label ambiguity rather than model capacity.
3) **Personalized within-subject** models show preliminary promise: 95.1% F1 (4-class, trial-aware) on a single OpenBCI subject provides initial evidence that EEG emotion recognition may be achievable when individual variability is eliminated, pending multi-subject replication.

We recommend that the EEG-BCI community adopt trial-aware evaluation (Tier 2 or stricter) as the minimum reporting standard. Studies should include both a Tier 0 baseline and Tier 2 result to transparently quantify the role of leakage in their reported performance.

Reproducibility

All code, trained models, and extracted features are available in our public repository. The evaluation pipeline uses fixed random seed (42) across NumPy, PyTorch, and scikit-learn. Total computational cost: feature extraction (3.6 h) + hyperparameter optimization (6.1 h) + full evaluation (0.5 h) on a single NVIDIA RTX GPU.

References

[1] S. Koelstra *et al.*, "DEAP: A database for emotion analysis using physiological signals," *IEEE Trans. Affective Computing*, vol. 3, no. 1, pp. 18–31, 2012.

[2] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Proc. ICLR*, 2018.

[3] P. Zhong, D. Wang, and C. Miao, "EEG-based emotion recognition using regularized graph neural networks," *IEEE Trans. Affective Computing*, vol. 11, no. 3, pp. 532–541, 2020.

[4] G. Varoquaux *et al.*, "Assessing and tuning brain decoders: Cross-validation, caveats, and guidelines," *NeuroImage*, vol. 145, pp. 166–179, 2017.

[5] S. Roberts *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017.

[6] S. Saeb, L. Lonini, A. Jayaraman, D. C. Mohr, and K. P. Kording, "The need to approximate the use-case in clinical machine learning," *GigaScience*, vol. 6, no. 5, pp. 1–9, 2017.

[7] W.-L. Zheng and B.-L. Lu, "Investigating critical frequency bands and channels for EEG-based emotion recognition with deep neural networks," *IEEE Trans. Autonomous Mental Development*, vol. 7, no. 3, pp. 162–175, 2015.

[8] A. Craik, Y. He, and J. L. Contreras-Vidal, "Deep learning for electroencephalogram (EEG) classification tasks: A review," *J. Neural Engineering*, vol. 16, no. 3, p. 031001, 2019.

[9] F. Lotte *et al.*, "A review of classification algorithms for EEG-based brain-computer interfaces: A 10 year update," *J. Neural Engineering*, vol. 15, no. 3, p. 031005, 2018.

[10] J.-P. Lachaux, E. Rodriguez, J. Martinerie, and F. J. Varela, "Measuring phase synchrony in brain signals," *Human Brain Mapping*, vol. 8, no. 4, pp. 194–208, 1999.

[11] R. T. Canolty and R. T. Knight, "The functional role of cross-frequency coupling," *Trends in Cognitive Sciences*, vol. 14, no. 11, pp. 506–515, 2010.

[12] L.-C. Shi, Y.-Y. Jiao, and B.-L. Lu, "Differential entropy feature for EEG-based emotion recognition," in *Proc. 7th Int. IEEE/EMBS Conf. Neural Eng.*, pp. 81–84, 2013.

[13] J. J. B. Allen, J. A. Coan, and M. Nazarian, "Issues and assumptions on the road from raw signals to metrics of frontal EEG asymmetry in emotion," *Biological Psychology*, vol. 67, pp. 183–218, 2004.